

PA-2

Instructions

- This assignment will count towards bonus marks. PA-1 and PA-2 together can fetch you at most 5/100 in the total score.
- This assignment is not going to affect your eligibility to write the end-term. That is only dependent on the graded assignments.
- You have to submit a report along with the code.
- Use this [form](#) to submit your report and code.
- There are three questions in this assignment.
- The deadline is 24th August, 2025, 23:59 PM.

Question-1: Spectral Clustering using NumPy

Consider the dataset $\mathbf{X}$ generated using the code given below. Using only the libraries NumPy and Matplotlib, do the following:

- Use the distance metric as $\exp\left(\frac{-||x - y||^2}{2\sigma^2}\right)$ and induce a weighted graph. Use $\sigma = 2$.
- Compute the weight matrix W , the diagonal matrix D and from this the Laplacian L .
- Find the three smallest eigenvalues of the Laplacian.
- Using the eigenvectors corresponding to the two smallest non-zero eigenvalues, construct 2-dimensional embeddings for all the data-points. Ensure that you normalize this embeddings to have unit norm. Visualize these embeddings by means of a scatter plot.
- Run Kmeans with $k = 3$ on this embedding and report your observations.

```

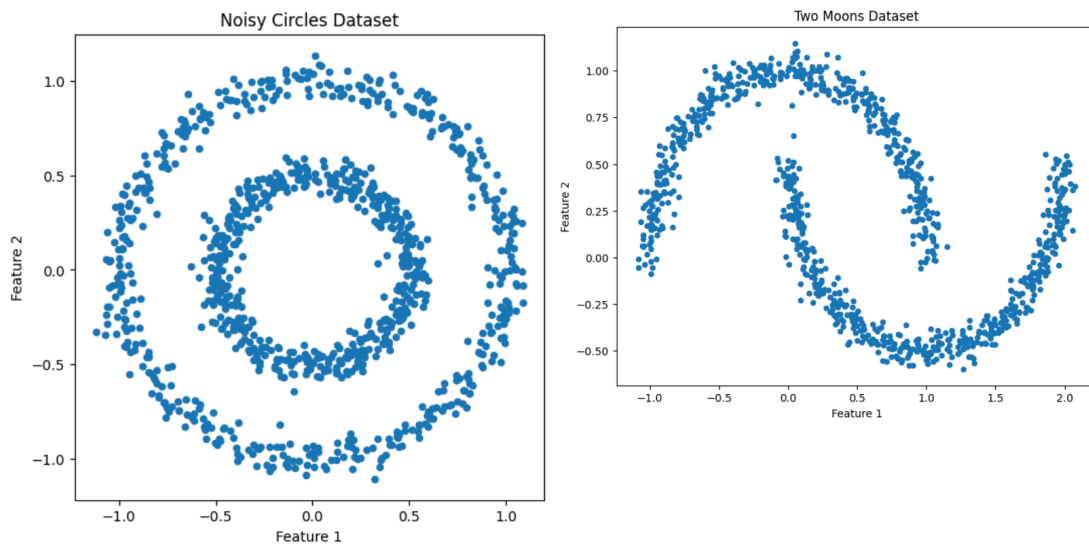
import numpy as np
import matplotlib.pyplot as plt
rng = np.random.default_rng(seed=1001)

x1 = rng.multivariate_normal(
    mean=[2, 2, 2], cov=np.eye(3), size=50
)
x2 = rng.multivariate_normal(
    mean=[-3, -3, -3], cov=np.eye(3), size=50
)
x3 = rng.multivariate_normal(
    mean=[-2, -2, 6], cov=np.eye(3), size=50
)
X = np.concatenate(
    [x1, x2, x3], axis=0
)

```

Question-2: Spectral Clustering on Toy Datasets

- Run both Kmeans and spectral clustering on the following datasets and compare the results. Use $k = 2$ for each dataset-algorithm pair.
- Use the sklearn APIs for both algorithms.



Question-3: Image Segmentation

Study the image segmentation literature to understand how spectral clustering can be used to segment images. Get hold of some image on the internet and segment it using spectral clustering. You can use the sklearn APIs [here](#). Comment on the results you obtain.